

STM32 Project Setup Report

User

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1 Introduction

This report details the steps taken to configure, generate code, and debug an STM32 project using STM32CubeMX, VisualGDB, and OpenOCD. The goal was to configure USART, GPIO, and basic system peripherals, generate the necessary initialization code, and set up debugging using ST-Link.

2 STM32CubeMX Configuration

1. USART Configuration

- Configured USART1 with the following parameters:
 - Baud Rate: 9600 (or 115200 depending on application)
 - Parity: None
 - Flow Control: Disabled
- Mapped PA9 as USART1_TX and PA10 as USART1_RX.

2. GPIO Configuration

- Configured PA5 as GPIO_Output for an LED.
- Default settings for GPIO: Output level low, Output Push-Pull, No pull-up/pull-down resistors, and low speed.

3. System Configuration (Sys)

- Enabled Serial Wire Debug (SWD) for debugging.

4. Clock Configuration

- Configured system clock using the HSE (High-Speed External) with a crystal oscillator.
- Set PLL to generate the required system clock frequency.

5. SysTick and Wake-Up

- SysTick enabled by default for system timing.
- Wake-up sources and low-power modes left at default.

6. Watchdog Timers

- Both IWDG and WWDG were left disabled for this development phase.

7. Additional Peripherals

- DMA was not enabled, as it was unnecessary for this basic setup.

3 VisualGDB Setup

1. Successfully integrated STM32CubeMX with VisualGDB.
2. Debugging method selected: `OpenOCD ST-Link Fork`.
3. Debug settings:
 - Debug output from OpenOCD: Enabled.
 - USB Serial Number passed to OpenOCD: Disabled (only one ST-Link in use).
 - GDB Commands used:

```
set remotetimeout 60
target remote :$$SYS:GDB_PORT$$
monitor arm semihosting enable
mon halt
load
mon reset init
```

4 Code Generation and Debugging

1. Code was successfully generated using STM32CubeMX with an `.ioc` file.
2. VisualGDB built the project, and the debugger hit the breakpoint.

5 Next Steps

The next steps involve resolving the issue with viewing hardware registers during debugging. The project is currently functional with breakpoints being hit, but hardware register visibility needs to be addressed.